

Vishal Samal

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EXPERIENCE

AI Scientist

May 2024 – Present

ConcertAI

Bengaluru, Karnataka

- Deployed core NLP models for our generative AI platform, leveraging ONNX and inference optimizations to **reduce inference resource usage by 25%**, enabling the system to process **15,320 patients and 1.97 million documents** efficiently.
- Developed and deployed deep learning models to extract cancer biomarkers from patient data, **achieving an 89.2% F1-score across 1,823 biomarkers** and 13 cancer types.
- Designed a transformer-based information extraction pipeline for processing patient notes, **increasing pipeline accuracy by 19.8%**.
- Built and deployed robust text classification models to standardize medical concepts and filter irrelevant entities, achieving an **average classification accuracy of 94.7% across multiple cancer datasets**.

Data Scientist

July 2022 – April 2024

Elucidata

New Delhi, Delhi

- Developed and deployed a data analysis agent with a chat-based interface to search and visualize datasets, leveraging text embedding models with Python and SQL tools. Enabled non-technical scientists to **execute complex queries and visualizations with 90.4% accuracy**.
- Led a team to fine-tune large language models (LLMs) for information extraction from preclinical trial research, **leveraging LoRA and distributed GPU training**. Trained 3B and 7B parameter models, **increasing performance by 35.8%**.
- Designed and implemented a **CI/CD** pipeline for automated deployment and testing of REST APIs for model inference, using **Docker, FastAPI, AWS ECR, and SageMaker**.

Data Analyst

June 2021 – July 2022

Elucidata

New Delhi, Delhi

- Contributed to the NLP module for **ingesting and curating about 40TB of data**, developing inference modules and functional tests for 12 medical entities.
- Developed a training pipeline for embedding models using contrastive learning, enhancing biomedical concept search and normalization, improving **standardization accuracy upto 92%** and **reducing entity loss to under 5%**.

EDUCATION

IIIT Delhi

New Delhi, Delhi

Bachelor of Technology Electronics and Communication Engineering

Aug. 2017 – June 2021

PUBLICATIONS

Reducing Inference Time of Biomedical NER Tasks using Multi-Task Learning

December 2022

- Proposed and developed a novel multi-task learning framework to jointly train two or more biomedical NER tasks, enhancing model efficiency while maintaining accuracy.
- Our experiments with three biomedical NER tasks resulted in a nearly threefold increase in inference time without any reduction in accuracy.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Bash, HCL

Frameworks: Pytorch, Keras, XGBoost, scikit-learn, FastAPI

Developer Tools: Git, Docker, AWS S3, AWS ECR, Sagemaker, CI/CD, Terraform

Libraries: Pandas, NumPy, Matplotlib, Seaborn